# Proposal for SAE level 4 Autonomous Mobility Solutions








TEAM
# AVERERA

"Driven by Innovation, Crafted with Precision"

# About Us

Team AVERERA is a group of automobile students at IIT (BHU) Varanasi dedicated to designing and developing the future of mobility. We are working on hyper-efficient EVs and urban concept autonomous vehicles. Related to autonomy, our prior projects include development of SAE level 3 autonomous test vehicle. The technology was then transferred to develop an autonomous golf cart. A commercial grade **SAE Level 4 autonomous vehicle** is our next frontier in urban mobility. The goal is to develop a Level-4-capable autonomous mobility platform capable of navigating complex campus environments featuring mixed traffic, dynamic obstacles, varying road conditions, and limited lane discipline. It should be capable of detecting and safely responding to vulnerable and unstructured road users commonly encountered in Indian environments, including pedestrians, cyclists, street animals, and low-speed campus traffic.

## Vision and Mission

Founded in 2013 and guided by the principle: “Leave the world better than you found it”: Our vision is to develop future-ready engineers capable of designing efficient, reliable, and sustainable transportation technologies.

Our mission is to:

- Promote innovation in autonomous mobility.
- Provide students with hands-on engineering experience, train students in industry-relevant autonomous vehicle engineering practices.
- Build strong collaborations between academia and industry.
- Contribute towards global autonomous transportation system.

# Contents

**Executive Summary** ..... 4

**Budget Summary** ..... 5

**Project Breakdown** ..... 6

**1. Drive-by-Wire (DBW) Conversion & Actuation Hardening** ..... 6

**2. Multi-Modal Perception & 360° Sensor Fusion** ..... 6

**3. Tactical Localization, Mapping & RTK-GNSS Navigation** ..... 7

**4. Industrial Compute, Motion Planning & AI Algorithms** ..... 8

**Long-Term Impact** ..... 10

**Risk Assessment & Mitigation** ..... 11

**Deliverables to Partners** ..... 12

**Previous Partners** ..... 13

**Conclusion** ..... 13

# Executive Summary

A commercial grade SAE Level 4 autonomous vehicle is our next frontier in urban mobility. This project directs towards transforming a standard golf cart into a futuristic autonomous urban pod. It should be capable of navigating complex campus environments featuring mixed traffic, dynamic obstacles, varying road conditions, and limited lane discipline. Keeping pedestrian safety at priority, we understand that whatever we develop should be capable of detecting and safely responding to vulnerable and unstructured road users commonly encountered in Indian environments, including pedestrians, cyclists, street animals, and low-speed campus traffic.

Beyond developing a vehicle, as part of the autonomous vehicle development program, Team AVERERA aims to pursue the following objectives, subject to technical feasibility, regulatory requirements, and institutional approvals, which we are committed to sharing non-confidential outputs with our partners too:

1. **Creation of a High-Definition Campus Mobility Map**
   Develop and periodically update a high-resolution digital map of the IIT (BHU) campus using data collected from the autonomous vehicle platform. The generated maps may be made available for educational, research, and non-commercial academic applications.
2. **Autonomous Mobility Dataset Generation**
   Build and curate a structured dataset of campus driving scenarios, including pedestrian interactions, animal crossings, intersections, and varying environmental conditions, to support future academic research and algorithm development.
3. **Establishment of an RTK-GNSS Infrastructure Testbed**
   Explore the deployment of an RTK-GNSS base station within the IIT (BHU) campus to support centimeter-level localization accuracy for autonomous systems research, subject to necessary institutional and regulatory approvals.
4. **Indigenous Perception and Localization Research**
   Investigate cost-effective sensor fusion techniques integrating cameras, GNSS, IMU, and other sensors to achieve reliable localization and environmental perception under diverse operating conditions.
5. **Digital Twin and Simulation Framework Development**
   Create a virtual simulation environment and digital twin of the campus to enable safe testing, validation, and optimization of autonomous driving algorithms.

The outputs of these can be made available to the partners or open to the public domain for research/educational activities depending on institute policies, and nature of agreement with our partner(s).
The project establishes permanent research infrastructure across five core engineering domains:

1. Drive-by-Wire (DBW) Conversion & Actuation Hardening
2. Multi-Modal Perception & 360° Sensor Fusion
3. Tactical Localization, Mapping & RTK-GNSS Navigation
4. Industrial Compute, Motion Planning & AI Algorithms

To eliminate single points of failure and ensure seamless subsystem integration, all R&D spans a 24-month period structured into a rigorous engineering roadmap. Each subtask has been divided into 4 or 5 phases, as per the nature of the task.

# Budget Summary

The total project net worth is valued at **₹54,28,000 (INR)**. The budget is strictly allocated to purchasing equipments for the project. The prices listed are latest to the best of our knowledge.

| S.No. | Research Vertical / Activity | Consumables & Fabrication (₹) | Permanent Assets & Equipment (₹) | Total Cost (₹) |
|---|---|---|---|---|
| **1** | Drive-by-Wire (DBW) Conversion and Actuation Hardening | 1,50,000 | 5,84,000 | 7,34,000 |
| **2** | Multi-Modal Perception & Vision | 80,000 | 27,60,000 | 28,40,000 |
| **3** | Tactical Localization & RTK-GNSS | 75,000 | 11,84,000 | 12,59,000 |
| **4** | Industrial Compute & AI Algorithms | 80,000 | 5,15,000 | 5,95,000 |
| **—** | **Grand Total** | **₹3,85,000** | **₹50,43,000** | **₹54,28,000** |

Companies can support by:

1. Financial sponsorship of full project/one or more R&D activity(es).
2. Sponsorship in kind by providing some parts/equipment or manufacturing the same for the team.
3. Technical mentorship: access to premium software used at the industry level for simulations; algotihms, datasets; lectures/seminars/webinars/curated learning material; design reviews/evaluations.
4. Logistics support: Fund our delivery, import/export charges.

# Project Breakdown

## 1. Drive-by-Wire (DBW) Conversion & Actuation Hardening

To operate autonomously with automotive reliability, the mechanical steering, braking, and propulsion systems of the base golf cart are converted to deterministic electronic interfaces controlled via CAN-bus commands.

**Why Drive-by-Wire Hardening Matters**

- Bridges computational trajectory planning with physical vehicle motion.
- Provides sub-10 ms deterministic actuation for steering angle and braking torque.
- Incorporates hardware and software fail-safes to permit instantaneous manual takeover.
- Serves as the foundational testbench for advanced vehicle dynamics and control algorithms.

**Phase-by-Phase Task Distribution**

- **Phase I (Months 1–4): Tear-Down & Actuation Hardening**
  - Strip the base golf cart down to its bare mechanical chassis.
  - Design structural mounting brackets and other needed equipments for proper safety of sensors. Make changes in the vehicle if needed.
  - Install the Electronic Power Steering (EPS) column and dual Kar-Tech linear hydraulic brake actuators.
  - Program low-level CAN bus control loops to ensure zero mechanical slop during remote steering and braking.
  - Integrate hardware emergency-stop relays and brake-override springs directly into the actuation chain.
- **Phase II (Months 5–11): Latency & Step-Response Calibration**
  - Benchmark steering rate-of-turn and hydraulic brake pressure response times against ROS 2 command timestamps.
- **Phase III (Months 12–18): Dynamic Actuator Tuning**
  - Tune low-level PID/MPC torque controllers across wet and dry pavement profiles.
- **Phase IV (Months 19–24): High-Speed Endurance Validation**
  - Validate actuator durability and emergency physical stopping distances during multi-hour continuous track testing.

**Cost Breakdown**

| Item / Real Market Equipment | Qty | Unit Price (₹) | Total Cost (₹) | Asset Classification |
|---|---|---|---|---|
| Electronic Power Steering (EPS) Column & Servo Actuator Kit | 1 | 1,85,000 | 1,85,000 | Permanent Asset |
| Linear Electro-Hydraulic Brake Actuators | 2 | 65,000 | 1,30,000 | Permanent Asset |
| Digital Throttle DAC & CAN-Bus Controller Interface Module | 1 | 45,000 | 45,000 | Permanent Asset |
| Dual-Channel CAN-to-USB Interfaces | 2 | 67,000 | 1,34,000 | Permanent Asset |
| Precision Steering Torque Sensor & Optical Rotary Angle Encoder | 1 | 90,000 | 90,000 | Permanent Asset |
| Miscellaneous (Machined CNC Brackets, Mechanical Linkages, Safety Relays, PCBs etc.) | — | 1,50,000 | 1,50,000 | Consumable |
| **Total Consumables & Fabrication** | | | **₹1,50,000** | |
| **Total Permanent Assets & Equipment** | | | **₹5,84,000** | |
| **Grand Subsystem Total** | | | **₹7,34,000** | |

## 2. Multi-Modal Perception & 360° Sensor Fusion

A Level 4 urban pod requires 360-degree redundancy capable of operating in dense urban traffic, varying lighting, and adverse weather. We combine a primary 3D digital LiDAR, short-range solid-state corner LiDARs and industrial RGB-D vision cameras.

**Why Perception & Sensor Fusion Matters**

- 3D Industrial Digital LiDAR: Delivers dense, 360-degree spatial point clouds for precise geometry and obstacle detection.
- Solid-State Corner LiDAR (Livox Mid-360): Eliminates blind spots around the wheels and front bumper.

- RGB-D Cameras (Intel RealSense / FLIR): Enables semantic segmentation (lane markings, pedestrian classification, traffic signs).
- 360° Sensor Fusion: Mitigates single-sensor failure modes by generating a unified, probabilistic bird's-eye-view (BEV) occupancy grid.

**Phase-by-Phase Task Distribution**

- **Phase I (Months 1–6): Sensor Procurement & Bench Testing**
  - Purchase all required sensors
  - Characterize baseline sensor noise, radar field-of-view (FoV), and camera intrinsic parameters.
- **Phase II (Months 9–13): Sensor Integration & Extrinsic Calibration**
  - Mount all LiDARs, and cameras using shielded cabling.
  - Time-synchronize all sensors via Precision Time Protocol (PTP) over industrial Ethernet.
  - Execute automated multi-modal extrinsic target calibration using checkerboards and retroreflective targets.
- **Phase III (Months 14–18): Perception Pipeline & Data Collection**
  - Harvest terabytes of real-world urban traffic data across diverse lighting and weather conditions.
  - Implement PointPillars / YOLOv8-3D (or likewise algorithms) for real-time 3D bounding box detection and Kalman filter fusion.
- **Phase IV (Months 19–24): Real-Time Edge Evaluation**
  - Validate tracking recall, false-positive rejection, and detection latency (<30 ms) on onboard AI computing modules.

**Cost Breakdown**

| **Item / Real Market Equipment** | **Qty** | **Unit Price (₹)** | **Total Cost (₹)** | **Asset Classification** |
|---|---|---|---|---|
| 3D Industrial Digital LiDAR Sensor (Ouster OS0/64 Series) | 1 | 17,20,000 | 17,20,000 | Permanent Asset |
| 3D Solid-State Perimeter LiDAR Sensors (Livox Mid-360) | 2 | 1,60,000 | 3,20,000 | Permanent Asset |
| Industrial RGB-D Vision Cameras | 4 | 1,80,000 | 7,20,000 | Permanent Asset |
| Miscellaneous (Optical Calibration Boards, Sensor Shrouds & Shielded Cabling, etc.) | — | 80,000 | 80,000 | Consumable |
| **Total Consumables & Fabrication** | | | **₹80,000** | |
| **Total Permanent Assets & Equipment** | | | **₹27,60,000** | |
| **Grand Subsystem Total** | | | **₹28,40,000** | |

## 3. Tactical Localization, Mapping & RTK-GNSS Navigation

To navigate urban layouts and competition courses reliably, the vehicle must determine its position with centimeter-level precision while resisting drift when satellite signals are occluded by trees or buildings.

**Why Tactical Localization Matters**

- Standard GPS exhibits error margins of 2–5 meters; RTK-GNSS achieves 2–3 cm accuracy.
- Dual-Antenna GNSS/INS (VectorNav VN-300) provides true heading drift-free orientation even when stationary.
- Fusing wheel odometry and LiDAR Simultaneous Localization and Mapping (SLAM) bridges satellite outages in urban canyons.

**Phase-by-Phase Task Distribution**

- **Phase I (Months 1–4): Base Station & Odometry Setup**
  - Install the Emlid Reach RS2+ base station on campus to broadcast RTK differential corrections via NTRIP/LoRa.
  - Mount optical rotary wheel encoders and characterize slip error across pavement types.
- **Phase II (Months 5–8): Inertial Sensor Placement**
  - Install the dual-antenna GNSS/INS at the vehicle's center of gravity with magnetic roof antenna mounts.
- **Phase III (Months 9–13): Localization Middleware Configuration**
  - Integrate Extended Kalman Filter (EKF) nodes combining IMU rates, wheel odometry, and RTK coordinates.

- **Phase IV (Months 14–18): HD Mapping & Semantic Annotation**
    - Generate High-Definition (HD) 3D point-cloud maps of target operating environments using LIO-SAM.
    - Build semantic vector layers (curbs, lane centerlines, stop lines, intersections) over the base maps.
- **Phase V (Months 19–24): GPS-Denied Drift Verification**
    - Simulate temporary GNSS dropouts to verify localization drift bounds (±5 cm) using LiDAR odometry and INS alone.

**Cost Breakdown**

| Item / Real Market Equipment | Qty | Unit Price (₹) | Total Cost (₹) | Asset Classification |
|---|---|---|---|---|
| Dual-Antenna RTK GNSS/INS System | 1 | 5,00,000 | 5,00,000 | Permanent Asset |
| Multi-Band RTK GNSS Base Station Kit | 1 | 6,60,000 | 6,60,000 | Permanent Asset |
| High-Precision Optical Rotary Wheel Encoders | 4 | 6,000 | 24,000 | Permanent Asset |
| Miscellaneous (GNSS Survey Poles, Magnetic Roof Mounts & Ground Truth Markers) | — | 75,000 | 75,000 | Consumable |
| **Total Consumables & Fabrication** | | | **₹75,000** | |
| **Total Permanent Assets & Equipment** | | | **₹11,84,000** | |
| **Grand Subsystem Total** | | | **₹12,59,000** | |

## 4. Industrial Compute, Motion Planning & AI Algorithms

The computational engine transforms perception occupancy grids and localization coordinates into collision-free trajectories executed in real time.

**Why Compute Infrastructure & AI Planning Matters**

- Autonomous driving requires deterministic, high-throughput processing with zero thermal throttling.
- NVIDIA Jetson AGX Orin (64GB): Provides up to 275 TOPS of edge-AI compute, running deep neural networks concurrently with ROS 2 navigation stacks.
- Motion planning must handle dynamic obstacle prediction, smooth velocity profiling, and curvature-constrained path generation.

**Phase-by-Phase Task Distribution**

- **Phase I (Months 1–6): Simulation Twin & Workstation Setup**
    - Build a high-fidelity digital twin of the vehicle in CARLA or Gazebo simulation environments.
    - Set up the simulation workstation for algorithm training and offline regression testing.
- **Phase II (Months 7–8): Compute Hardware Ruggedization**
    - Configure the dual NVIDIA Jetson AGX Orin edge computers inside an actively cooled, vibration-proof enclosure.
- **Phase III (Months 9–13): Middleware & Network Architecture**
    - Containerize ROS 2 nodes (Docker) and configure the industrial automotive Ethernet switch for real-time IPC.
- **Phase IV (Months 14–18): Motion Planning & Control Algorithm Development**
    - Develop a path planner for global route generation across HD road networks.
    - Implement a Model Predictive Controller (MPC) for local trajectory execution, accounting for tire friction and actuator lag.
- **Phase V (Months 19–24): Closed-Loop Edge Optimization**
    - Deploy algorithms to onboard hardware, implementing CPU/GPU thread scheduling to guarantee sub-50 ms end-to-end control latency.

**Cost Breakdown**

| Item / Real Market Equipment | Qty | Unit Price (₹) | Total Cost (₹) | Asset Classification |
|---|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| AI Edge Supercomputer Dev Kit (NVIDIA Jetson AGX Orin 64GB) | 1 | 4,00,000 | 4,00,000 | Permanent Asset |
| Rugged Industrial Automotive Ethernet Switch & Vehicle Hub | 1 | 1,15,000 | 1,15,000 | Permanent Asset |
| Storage NVMe SSDs, Custom Shielded Harnesses & Benchtop Power | 1 set | 80,000 | 80,000 | Consumable |
| **Total Consumables & Fabrication** | | | **₹80,000** | |
| **Total Permanent Assets & Equipment** | | | **₹5,15,000** | |
| **Grand Subsystem Total** | | | **₹5,95,000** | |

# Long-Term Impact

- **Permanent Level 4 Autonomous Research Platform:** Transforms a standard electric golf cart into a commercial-grade, multi-sensor autonomous testbench at the institute, serving researchers in robotics, applied AI, and control theory for years to come.
- **Indigenous Industry-Ready Competency:** Equips students with hands-on expertise in automotive-grade drive-by-wire systems, industrial sensor calibration, ROS 2 middleware, RTK-GNSS surveying, and ISO 26262 functional safety architectures.
- **Advanced Interdisciplinary Collaboration:** Unites undergraduate researchers across Mechanical, Electrical/Electronics, Computer Science, and Material Science Engineering around an integrated systems-engineering challenge.
- **Corporate Collaboration & Testbed Integration:** Establishes an outdoor mobility platform where sponsoring industry partners can benchmark novel sensors, compute platforms, and control algorithms in complex Indian urban environments.
- **High-Value Academic Output:** Generates rich, real-world multi-modal datasets (unstructured traffic, dynamic pedestrian interactions) that support peer-reviewed technical publications, master's theses, and open-source robotics contributions.

# Risk Assessment & Mitigation

Team AVERERA will adopt an iterative **"Design → Simulate → Prototype → Validate → Integrate"** development methodology to minimize technical and project risks. Laboratory-scale experimentation will be used extensively before committing resources to full-scale manufacturing, ensuring that only validated concepts are incorporated into the final vehicle. Regular design reviews, milestone tracking, and documentation practices will further ensure that project objectives are achieved within the planned timeline and budget. Yet, there remain chances of failure (as in any other research).

| Risk Category | Potential Risk | Mitigation Strategy |
|---|---|---|
| Technical Development | Experimental concepts may not achieve the expected performance. | Multiple concepts will be evaluated in parallel. Proven baseline designs will be retained as fallback options for vehicle integration. |
| Research Uncertainty | Simulation results may not correlate perfectly with experimental outcomes. | Experimental validation will be conducted at each stage to verify models and refine design assumptions before vehicle implementation. |
| Procurement & Supply Chain | Delays in procurement of sensors, electronics, materials, or custom-manufactured parts. | Procurement activities will begin early in the project timeline, with multiple vendor options identified for critical components. |
| Budget Constraints | Increase in material or equipment costs due to market fluctuations. | Budget estimates include contingency provisions and preference will be given to in-house manufacturing and institute facilities wherever feasible. |
| Equipment Availability | Limited availability of testing facilities, workshop resources, or laboratory equipment. | Testing schedules will be planned in advance and alternative institute facilities will be identified for critical activities. |
| Vehicle Integration | Individual subsystems may perform well independently but face compatibility challenges during vehicle integration. | Incremental subsystem integration and validation will be conducted throughout development rather than waiting until final assembly. |
| Data Acquisition & Testing | Sensor failures, inaccurate measurements, or incomplete datasets may affect research outcomes. | Calibration procedures, redundant measurements, and periodic verification of instrumentation will be implemented. |
| Timeline Management | Delays in research, fabrication, or testing activities may impact project completion. Further, examination schedules, internships or placements may hinder the performance of team individuals. | Project milestones will be monitored regularly, with buffer periods incorporated into the development schedule. |
| Team Continuity | Student graduation and turnover may result in knowledge loss during the project lifecycle. | Comprehensive documentation, design repositories, and knowledge transfer sessions will be maintained throughout the project. |

The project primarily involves moderate technical and execution risks typical of research and development activities. However, these risks are mitigated through phased development, extensive prototype validation, utilization of existing institute infrastructure, and the availability of proven fallback solutions.

# Deliverables to Partners

Team AVERERA views this as a collaborative partnership that enables innovation in sustainable mobility. It is committed to:

**Brand Visibility & Recognition**

- Sponsor logo placement on the Autonomous Vehicle.
- Logo placement on team apparel, pit setup, and promotional materials.
- Dedicated acknowledgment on Team AVERERA's website and social media platforms.
- Recognition during technical exhibitions, demonstrations, and institute events.

**Meaningful Technical Collaboration**

- Opportunity to engage with students working on advanced EV technologies.
- Access to non-confidential project outcomes, performance data, and technical findings.
- Opportunity to propose industry-relevant engineering challenges for student exploration.

**Talent & Recruitment Access**

- Direct interaction with highly motivated students possessing hands-on experience in electric vehicle development.
- Opportunities for internships, mentorship programs, and recruitment engagement.
- Early access to a multidisciplinary talent pool with expertise in design, simulation, manufacturing, electronics, controls, and testing.

**Institutional & Outreach Benefits**

- Recognition as a contributor to sustainable mobility and student innovation.
- Visibility across IIT (BHU)'s academic and student community.
- Association with a nationally and internationally recognized engineering competition.

**Project Deliverables**

Upon completion of the project, partners may be provided with:

- Project completion and non-confidential utilization reports.
- High-level technical summaries of research activities.
- Vehicle performance highlights.
- Documentation of infrastructure and facilities developed through the project.
- Impact assessment demonstrating the utilization of sponsored resources.

## Previous Partners

Team AVERERA has actively collaborated with many companies/startups in the past. Team AVERERA boasts of its alumni in multiple distinguished companies, which we actively keep leveraging for collaboration. Over the past decade, we have had the fortune to partner with companies like Tata Motors, GoPro, HP. Nike, MATLAB, drobot and many more.




## Conclusion

This project goes beyond mere competitions. The requested support will directly enable the creation of research infrastructure, experimental facilities, prototype development programs, and the final competition vehicle. More importantly, it will establish a long-term foundation for innovation in electric mobility, energy-efficient transportation, and sustainable engineering at IIT (BHU) Varanasi. By supporting Team AVERERA, partners will not only contribute to the development of a world-class energy-efficient vehicle but also help cultivate the next generation of engineers working toward India's clean mobility future.